

Roll No. 230162

Total No. of Pages: 02

Integrated M.C.A. (Semester -VII)
Examination Dec. - 2025
IMCA - 701T
Informaion and Network Security System

Time Allowed: Three Hours

Maximum Marks: 70
Min. Marks: 28

This question paper contains two sections as under

Section - A

Max. Marks-20

This section will have 10 questions, having 02 questions from each unit with no internal choice. Answer in about 20 words. All questions will have equal marks. (10×02=20)

Section - B

Max. Marks-50

This section will have 10 questions, having 02 questions from each unit. **One question to be attempted from each unit.** Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks. (10×05=50)

Section - A

- I. (i) What is the main goal of network Security ?
- (ii) Differentiate between threat and attack.
- (iii) Give name of one algorithm each for symmetric and asymmetric encryption.
- (iv) Encrypt P = "HELLO" using Caesar cipher if $k = 3$.
- (v) When one participant in a communication pretends to be someone else, it is called _____.
 - a. Passive attack
 - b. Virus attack
 - b. Masquerade
 - c. Data Driven attack
- (vi) The procedure to add bits to the last block is termed as _____.
- (vii) Full form of PKI is _____.
- (viii) Calculate & list $\phi(35)$.
- (ix) There are two modes of IP Sec : (a) _____ and (b) _____.
- (x) Define firewall.

Section B

Unit- 1

11. Discuss and distinguish active and passive attacks. Give example for each type of attack. What do you mean by Brute force attack?

Discuss :

- (a) Confidentiality
- (b) Integrity
- (c) Authentication
- (c) Availability

Unit-II

- (III) (a) What are the essential ingredients of a symmetric cipher ? Draw a diagram to explain stream & block ciphers. 6
- (b) What is the purpose of S-boxes in DES cipher ? 4

OR

Describe the four stages used in AES.

Unit-III

- (IV) (a) Explain the purpose of Hash function. (4)
- (b) Discuss how does confidentiality & integrity achieved by hash function. Use appropriate diagrams. 6

OR

- (a) List any two properties a digital signature should essentially have ? 4
- (b) Discuss – (a) Birthday Paradox
- (b) Digital signature standard

Unit - IV

- (V) (a) What are the main components of public key cryptography ? Discuss & draw the model for public key cryptography. 6
- (b) State and explain fermat's theorem. 4

Or

- (a) Discuss different steps of Diffie Hellman key exchange algorithm. 5
- (b) How are the public and private keys decided or generated for the RSA algorithm ? Write steps. 5

Unit – V

- (VI) Why the security of an e-mail is important ? How dos PGP help to provide e-mail security. 046

OR

Write note on –

- (a) Intrusion Detection system
- (b) Packet filter firewall
- (c) Cyber crimes